

Theme 1 Definitions

Saturday, 08 June 2024 01:37

$$\begin{aligned} |a| &= a && \text{if } a \geq 0 \\ |a| &= -a && \text{if } a < 0 \end{aligned}$$

$$\sqrt{a^2} = |a|$$

5 Properties of Absolute Values Suppose a and b are any real numbers and n is an integer. Then

$$\begin{array}{lll} 1. |ab| = |a||b| & 2. \left| \frac{a}{b} \right| = \frac{|a|}{|b|} \quad (b \neq 0) & 3. |a^n| = |a|^n \end{array}$$

6 Suppose $a > 0$. Then

4. $|x| = a$ if and only if $x = \pm a$
5. $|x| < a$ if and only if $-a < x < a$
6. $|x| > a$ if and only if $x > a$ or $x < -a$

The Vertical Line Test A curve in the xy -plane is the graph of a function of x if and only if no vertical line intersects the curve more than once.

■ Even and Odd Functions

If a function f satisfies $f(-x) = f(x)$ for every number x in its domain, then f is called an **even function**. For instance, the function $f(x) = x^2$ is even because

$$f(-x) = (-x)^2 = x^2 = f(x)$$

The geometric significance of an even function is that its graph is symmetric with respect to the y -axis (see Figure 19). This means that if we have plotted the graph of f for $x \geq 0$, we obtain the entire graph simply by reflecting this portion about the y -axis.

If f satisfies $f(-x) = -f(x)$ for every number x in its domain, then f is called an **odd function**. For example, the function $f(x) = x^3$ is odd because

$$f(-x) = (-x)^3 = -x^3 = -f(x)$$

The graph of an odd function is symmetric about the origin (see Figure 20). If we already have the graph of f for $x \geq 0$, we can obtain the entire graph by rotating this portion through 180° about the origin.

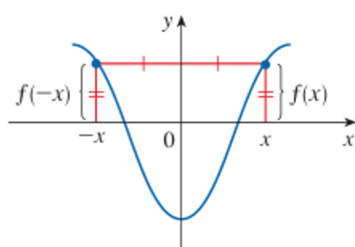


FIGURE 19
An even function

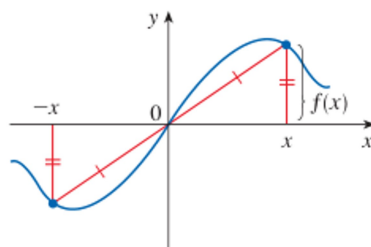


FIGURE 20
An odd function

A function f is called **increasing** on an interval I if

$$f(x_1) < f(x_2) \quad \text{whenever } x_1 < x_2 \text{ in } I$$

It is called **decreasing** on I if

$$f(x_1) > f(x_2) \quad \text{whenever } x_1 < x_2 \text{ in } I$$

$$\pi \text{ rad} = 180^\circ$$

$$\begin{array}{ll} \sin \theta = \frac{\text{opp}}{\text{hyp}} & \csc \theta = \frac{\text{hyp}}{\text{opp}} \\ \cos \theta = \frac{\text{adj}}{\text{hyp}} & \sec \theta = \frac{\text{hyp}}{\text{adj}} \\ \tan \theta = \frac{\text{opp}}{\text{adj}} & \cot \theta = \frac{\text{adj}}{\text{opp}} \end{array}$$

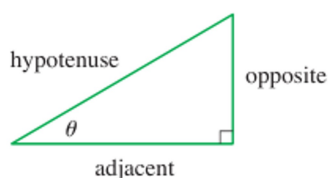


FIGURE 6

1 Definitio A function f is called a **one-to-one function** if it never takes on the same value twice; that is,

$$f(x_1) \neq f(x_2) \quad \text{whenever } x_1 \neq x_2$$

The **definition** of a one-to-one function can be written in two ways:

A function $f : A \rightarrow B$ is one-to-one if, for all $x_1, x_2 \in A$, $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$.

We can also say: A function $f : A \rightarrow B$ is one-to-one if $f(x_1) = f(x_2) \implies x_1 = x_2$.

Horizontal Line Test A function is one-to-one if and only if no horizontal line intersects its graph more than once.

2 Definitio Let f be a one-to-one function with domain A and range B . Then its **inverse function** f^{-1} has domain B and range A and is defined by

$$f^{-1}(y) = x \iff f(x) = y$$

for any y in B .

domain of $f^{-1} = \text{range of } f$

range of $f^{-1} = \text{domain of } f$

$f^{-1}(f(x)) = x$ for every x in A

$f(f^{-1}(x)) = x$ for every x in B

The graph of f^{-1} is obtained by reflecting the graph of f about the line $y = x$.

$$\log_b x = y \iff b^y = x$$

$$\begin{aligned}\log_b(b^x) &= x \quad \text{for every } x \in \mathbb{R} \\ b^{\log_b x} &= x \quad \text{for every } x > 0\end{aligned}$$

$$\log_e x = \ln x$$

$$\ln x = y \iff e^y = x$$

$$\begin{aligned}\ln(e^x) &= x \quad x \in \mathbb{R} \\ e^{\ln x} &= x \quad x > 0\end{aligned}$$

Laws of Logarithms If x and y are positive numbers, then

1. $\log_b(xy) = \log_b x + \log_b y$

2. $\log_b\left(\frac{x}{y}\right) = \log_b x - \log_b y$

3. $\log_b(x^r) = r \log_b x$ (where r is any real number)

11 Change of Base Formula For any positive number b ($b \neq 1$), we have

$$\log_b x = \frac{\ln x}{\ln b}$$

$$\sin^{-1}x = y \iff \sin y = x \quad \text{and} \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

$$\cos^{-1}x = y \iff \cos y = x \quad \text{and} \quad 0 \leq y \leq \pi$$

$$\tan^{-1}x = y \iff \tan y = x \quad \text{and} \quad -\frac{\pi}{2} < y < \frac{\pi}{2}$$